

Assignment: NET Centric Computing

Course Code: CSC378

Topic: Unit 2 - Introduction to ASP.NET (Theory & Practical)

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Due Date: 24 August

Submission Instructions:

Please complete the following questions by combining your theoretical explanations with practical implementations. For the practical components, include the relevant commands, code snippets, and screenshots of your terminal/console output to verify your work.

Questions:

1. Frameworks & CLI Initialization

Explain about .NET, .NET Core, and Mono frameworks. Using the .NET Command-Line Interface (CLI), generate a new ASP.NET Core Web API project. Execute the specific commands required to create (dotnet new), compile (dotnet build), and execute (dotnet run) the application. Verify that the application runs successfully by accessing the local port provided in your terminal output and include a screenshot.

2. Compilation, MSIL, and CLR

Explain the compilation and execution process of .NET applications covering the CLI, MSIL, and CLR. Write a basic C# Console Application, compile it using the dotnet build command, and demonstrate (with file path/screenshot) how to locate the generated MSIL .dll file in the build output directory to show what the CLR will execute.

3. Architecture & Deployment

Describe the .NET architecture and the procedure for deploying a .NET Core application. Create a basic ASP.NET Core application and use the dotnet publish -c Release command to package the application for production deployment. Compare the files generated in the standard build output directory versus the published output directory, and explain the differences.

4. Web Forms & Automated Testing

What is the purpose of ASP.NET Web Forms, and how does the modern ASP.NET Core framework handle testing differently? Create a .NET Core solution containing a C# Class Library and an xUnit Test project. Use the CLI to link the test project to the class library (dotnet add reference), write a basic assertion test, and execute the test suite using the dotnet test command. Document the terminal output of your test results.